

CS 556 Homework - Derivatives

Solve the following questions. Type your solutions using L^AT_EX or any other program you prefer and submit your typed solutions as a PDF file.

Question 1 (20 points) Use the limit definition of the derivative to exactly evaluate the derivative:

1. $f(x) = \sqrt{x+4}$

2. $f(x) = \frac{3}{x}$

Question 2 (20 points) Find the derivatives of the following functions:

1. $f(x) = 3x^3 - \frac{4}{x^2}$

2. $f(x) = (4 - x^2)^3$

3. $f(x) = e^{\sin(x)}$

4. $f(x) = \ln(x+2)$

5. $f(x) = x^2 \cos(x) + x \tan(x)$

6. $f(x) = \sqrt{3x^2 + 2}$

7. $f(x) = \frac{x}{4} \sin^{-1}(x)$

8. $x^2y = (y+2) + xy \sin(x)$

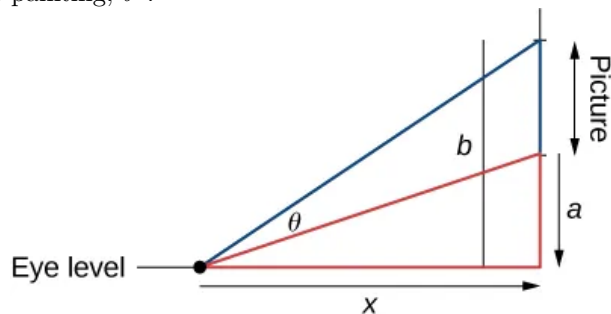
Question 3 (20 points) The following questions consider the wind speeds of Hurricane Katrina, which affected New Orleans, Louisiana, in August 2005. The data are displayed in a table.

Hours After Midnight, August 26	Wind Speed (mph)
1	45
5	75
11	100
29	115
49	145
58	175
73	155
81	125
85	95
107	35

a) Using the table, estimate the derivative of the wind speed at hour 39. What is the physical meaning?

b) Estimate the derivative of the wind speed at hour 83. What is the physical meaning?

Question 4 (20 points) The famous Regiomontanus' problem for angle maximization was proposed during the 15th century. A painting hangs on a wall with the bottom of the painting a distance a feet above eye level, and the top b feet above eye level. What distance x (in feet) from the wall should the viewer stand to maximize the angle subtended by the painting, θ ?



Question 5 (20 points) An airline sells tickets from Tokyo to Detroit for \$1200. There are 500 seats available and a typical flight books 350 seats. For every \$10 decrease in price, the airline observes an additional five seats sold. What should the fare be to maximize profit? How many passengers would be onboard?